

High-Quality Surface-Based 3D Reconstruction Using 2.5D Maps

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ABSTRACT

Previous works on RGB-D reconstruction are based on voxels, points, or meshes, which are either too computationally expensive to represent high-resolution 3D models, or not convenient for model regularization to deal with noise and errors. In this paper, we propose a new method to reconstruct 3D models with more accurate geometric information and better texture. Our method uses abstract surfaces consisting of different points with similar information as units of the model. To reduce the complexity, we use 2.5D heightmaps to represent each surface in the reconstructed model, making it convenient to do regularization. Experiments demonstrate the effectiveness of our method.

Index Terms: Human-centered computing—Human computer interaction—Virtual reality; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

3D reconstruction refers to establishing a mathematical model of 3D scenes or objects from 2D images. As the RGB-D sensor devices like Microsoft Kinect have evolved quickly during the last decade, RGB-D reconstruction has achieved significant development. A high-quality RGB-D reconstruction model must meet two criteria: correct geometric information and high-quality texture. Many works have been proposed to improve the quality of the model.

There are three ways to describe and construct RGB-D reconstructed models: Voxels, Surfels, and Mesh. Voxel based method [5, 7, 9] use a set of voxel volumes with weight to decide where the model's surfaces are. They use the Truncated Signed Distance Field algorithm (TSDF) to get the weight of each voxel. They are too computationally expensive to run in real-time and takes too much memory and have low resolution. Another kind of works make 3D constructed model with separate points like surfel. Based on typical point-based work [4], Elasticfusion [8] uses surfel containing a point's radius, position, color, and normal information to describe the model and for dense frame-to-model tracking. Due to the strong independence between different surfels, the final model it makes is relatively discrete. Model's surfaces may not be as accurate as what we expect regarding the fidelity of geometry, which may cause loss of geometric information we want and then make holes in the model. Some 3D reconstruction works refine the model directly on the step of triangular meshing. Surfelmeshing [6] first refine depth maps to make surfel cloud smoother. Then, it uses meshing and remeshing to reorganize the triangular mesh. Because the triangle meshing does not consider the model's overall structure, the reconstructed result may lose geometric information of the real scene.

To solve the geometry and texture loss of the reconstructed model, we consider using "surfaces" to represent the 3D real scene. As we

know, surfaces are smoother than a set of points or 3D cubes and contain more model structure information, ensuring that the model it makes can have good geometric equality and high resolution. The surface contains abundant semantic information, which can do further optimization using semantic Simultaneous Localization and Mapping (SLAM) and is more stable than separate points.

Based on this motivation, we propose a new format of the model unit. Instead of using 3D voxels or separate points, we make groups of abstract surfaces with high resolution to represent the 3D real model. The surface is our basic unit of the reconstructed model, consisting of 3D points with similar position and normal. We store the surfaces in the format of 2.5D images to reduce 3D point cloud data complexity. In our work, we update the surfaces from the image of every frame. Our method can find which surface the points should be in by the similarity of the points and surface's points. Then we update the points to the model's surface for every point from the back-projection of the depth images. The surface can be extended, ensuring that points will not get lost, but it has its border. Because we can not use only one 2.5D map to represent the complicating surface of a real 3D scene, we may divide some abstract surfaces of the 3D scene into small surface groups, each stored in a 2.5D map. The model made by surface groups can get good geometric information and looks close to real 3D model as for texture.

2 METHOD

2.1 Overview

First, we update the abstract surfaces incrementally from RGB images and depth images from a moving RGB-D sensor. 3D points made by the back-projection of frame's depth images should be on the specific existing model's surface judged by the similarity of normals and position between frame points and surface. If we can not find a suitable surface, we will make a new model surface. Each of our model surfaces is stored in a 400×400 2.5D heightmap and its corresponding RGBmap got from RGB images from sensors. These surface groups make up the whole 3D reconstructed model.

Next, we perform the pose estimation using a frame-to-model strategy. Instead of only aligning depth maps between frame and model for geometric tracking, inspired by ElasticFusion [8], we additionally use a full color of the surface groups model to perform photometric and depth frame-to-model pose estimation. In a word, we will use RGB and depth maps to do tracking like what ElasticFusion [8] does.

2.2 Surfaces Updating

The target of this section is to make the 3D reconstructed model M , which consists of different surfaces S_i ($i \in N^+$). Each surface has its reference plane with central point and central point's normal as plane's normal. The reference plane makes up the reference pixel plane of the surface's 2.5D height map. For every sequential frame, we traverse each pixel of the depth map and the corresponding color map to get P_{ug} , n_{ug} and C_u to find which surface it should be on.

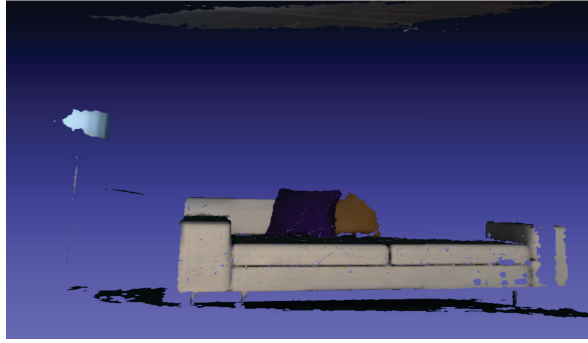
Once we project other points on the surface to the reference plane, their coordinate on the reference plane and height can be obtained. Then for every surface, we will calculate distance from the point P_{ug} to plane S_i and angle between n_{ug} and n_{si} ($i \in N^+$) to measure the

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(a) ElasticFusion



(b) Our Method

Figure 1: One of the ICL-NUIM's indoor scenes, left picture is ElasticFusion's global model result til frame 100 while right picture represent our method's 3D reconstruction result til frame 50.

similarity of point and surface. If there are no surfaces in our model or there is no suitable surface for the current 3D point from the pixel, we will make a new surface with its pose, central point, and normal. The central point is the 3D point itself, and the normal is also from the current point. Then we will find the best surface on which point P_{ug} should be. If there is no suitable surface for it in our model, we will use the point itself to make a new surface.

2.3 Pose Estimation

Given the 2.5D height maps, color maps of surfaces and pose of each surface got from **Surfaces Updating** procedure, we can restore them to the 3D point cloud of the scene in the world coordinate system using coordinate transformation. Then, based on **Iterative Closest Point** (ICP) [2] algorithm, we can align the predicted scene model's point cloud from previous frames and current frame's point cloud to predict current camera's pose. Similar to ElasticFusion, our pose estimation considers depth maps for frame-to-model tracking and full-color maps to do pose estimation.

3 EXPERIMENTS

We use the famous RGB-D dataset: ICL-NUIM [3] dataset and ElasticFusion authors' own lab dataset as our system's input. These datasets consist of RGB images and depth images with the resolution of 480×640 from RGB-D sensors like Kinect, Xtion etc. We use Ball-Pivoting [1] meshing algorithm on both our method and ElasticFusion to get the visualization results and compare geometry and texture performance between them. All experiments are performed on Ubuntu 16.04 with a GeForce GTX TITAN GPU card and an Inter(R) Core(TM) i5-4430 CPU. The result is shown on **Fig.1**.

From **Fig.1**, we can see that our method can reconstruct a better scene model in a relatively short time than ElasticFusion does. Our method's model contains more detailed information of the real scene such as floor lamp, sofa, plant and floor. Because our surface unit can store more structure information than separate surfel points, we can make 3D reconstruction model with more complete geometric information of real scene than surfel-based ElasticFusion's model. However, due to the large amount of data processed in each frame, this method is relatively slow. What's more, some surfaces may overlap sometimes, which needs subsequent improvements.

4 CONCLUSION

In this paper, we propose a surface-based 3D scene reconstruction. Instead of using fixed voxels or separate surfel clouds to represent the real model, we consider the surface as the unit of the reconstructed model, for the surface can contain more scene's whole structure information, which makes the 3D model more stable. To reduce the data complexity of the 3D model's point, we store the surface unit of the model in a 2.5D heightmap and corresponding RGB map. As is known that 2.5D map can not represent complex surface, we should use simple surfaces to make a full model surface. Experiments show that our surface-based method can perform better on geometric integrity compared with surfel-based modeling method and perform higher solution texture with little blur. As we can see, surface-based 3D models look better than some 3D reconstruction works both in geometry and texture.

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